

Lecture 4: Physical Layer - Wireless Communications

EC 441 - Introduction to Computer Networking

Boston University, Spring 2026

Learning Objectives

By the end of this lecture, you should be able to:

1. Describe basic wireless transmitter and receiver architectures
2. Calculate path loss using free-space and empirical models
3. Perform wireless link budget calculations
4. Explain how modulation enables channel selection and sharing
5. Understand QAM (quadrature amplitude modulation) and constellation diagrams
6. Identify spectrum allocations for WiFi, cellular, and other wireless services
7. Analyze cellular frequency reuse patterns
8. Understand interference and capacity limits in wireless networks

1 Wireless Transmitter and Receiver Architectures

1.1 Why Wireless is Different from Wired

Key Differences:

- **Broadcast medium:** Everyone can hear your transmission
- **Path loss:** Signal strength decreases rapidly with distance
- **Interference:** Multiple transmitters share the spectrum
- **Fading:** Signal varies due to multipath, obstacles
- **Regulation:** Spectrum is licensed or restricted
- **Power constraints:** Battery-powered devices

Benefits:

- **Mobility:** No physical connection needed
- **Deployment:** No cable installation
- **Flexibility:** Easy to add/remove devices

1.2 Basic Transmitter Architecture

Signal Flow:

[Data] → [Encoder] → [Modulator] → [Upconverter] → [PA] → [Antenna]

Key Components:

1. Encoder:

- Error correction coding (FEC)
- Interleaving
- Scrambling

2. Modulator (Baseband):

- Maps bits to symbols
- Pulse shaping (e.g., raised cosine)
- Generates I (in-phase) and Q (quadrature) signals

3. Upconverter (Mixer):

- Translates baseband to RF carrier frequency
- **Upconversion formula:**

$$s(t) = I(t) \cos(2\pi f_c t) - Q(t) \sin(2\pi f_c t) \quad (1)$$

where f_c = carrier frequency

4. Power Amplifier (PA):

- Amplifies signal to transmit power
- **Critical tradeoff:** Power vs. linearity vs. efficiency
- Typical efficiency: 20-40% (rest becomes heat!)

5. Antenna:

- Converts electrical signal to electromagnetic waves
- Directional or omnidirectional
- Gain measured in dBi (dB relative to isotropic)

1.3 Basic Receiver Architecture

Signal Flow:

[Antenna] → [LNA] → [Downconverter] → [Demodulator] → [Decoder] → [Data]

Key Components:

1. Low-Noise Amplifier (LNA):

- **Most critical component!**
- Amplifies weak received signal
- Minimizes noise figure (NF) - typically 1-3 dB
- High gain (15-25 dB) → subsequent stages less critical (Friis formula)

2. Downconverter (Mixer + LO):

- Translates RF to intermediate frequency (IF) or baseband
- Local oscillator (LO) must be phase-locked
- May use multiple stages (superheterodyne)

3. Demodulator:

- Extracts I and Q signals
- Matched filtering
- Symbol timing recovery
- Carrier phase recovery

4. Decoder:

- Error correction decoding
- De-interleaving
- Descrambling

1.4 Antenna Fundamentals

Isotropic Antenna (Theoretical):

- Radiates equally in all directions (sphere)
- Reference for antenna gain (0 dBi)
- **Doesn't exist in practice!**

Antenna Gain (G):

- Directivity: concentration of power in preferred direction
- Measured in dBi (dB relative to isotropic)

- **Omnidirectional** (e.g., dipole): ~ 2 dBi
- **Directional** (e.g., Yagi): 6-20 dBi
- **Dish** (e.g., satellite): 30-50 dBi

Key Property: Reciprocity

- Same gain for transmit and receive
- $G_{\text{tx}} = G_{\text{rx}}$ for same antenna

Effective Aperture:

$$A_e = \frac{G\lambda^2}{4\pi} \quad (2)$$

where λ = wavelength

Polarization:

- Vertical, horizontal, circular
- Mismatch causes 20+ dB loss!

2 Propagation Models

2.1 Free Space Path Loss (FSPL)

Ideal propagation: No obstacles, line-of-sight

Received power (Friis transmission equation):

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2 \quad (3)$$

where:

- P_t = transmit power
- G_t, G_r = transmit and receive antenna gains (linear)
- λ = wavelength = c/f
- d = distance

Path Loss in dB:

$$L_{\text{FSPL}} = 20 \log_{10}(d) + 20 \log_{10}(f) + 20 \log_{10} \left(\frac{4\pi}{c} \right) \quad (4)$$

Simplified (with f in MHz, d in km):

$$L_{\text{FSPL}} \text{ (dB)} = 32.45 + 20 \log_{10}(f_{\text{MHz}}) + 20 \log_{10}(d_{\text{km}}) \quad (5)$$

Key Insight: Path loss increases with:

- Distance: **20 dB per decade** ($10 \times$ distance $\rightarrow$ 20 dB more loss)

- Frequency: 20 dB per decade ($10\times$ frequency $\rightarrow$ 20 dB more loss)

Why higher frequencies lose more power:

- **Same transmit power** spread over larger area at receiver
- Effective antenna aperture $A_e \propto \lambda^2$ (shrinks with frequency)
- Not due to absorption in air (free space has no medium!)

WiFi Link

Parameters:

- Frequency: 2.4 GHz
- Distance: 100 m = 0.1 km
- Transmit power: 20 dBm (100 mW)
- Antenna gains: $G_t = G_r = 2$ dBi

Calculation:

$$\begin{aligned} L_{\text{FSPL}} &= 32.45 + 20 \log_{10}(2400) + 20 \log_{10}(0.1) \\ &= 32.45 + 67.6 - 20 = 80 \text{ dB} \end{aligned}$$

$$P_r = P_t + G_t + G_r - L = 20 + 2 + 2 - 80 = -56 \text{ dBm}$$

This is well above typical WiFi sensitivity (-80 to -90 dBm) $\rightarrow$ good link!

2.2 Beyond Free Space: Real-World Propagation

Real environments have:

- Obstacles (buildings, trees, walls)
- Reflections (ground, buildings)
- Diffraction (around corners)
- Scattering (small objects, rain)

Empirical Path Loss Models:

General form:

$$L(d) = L_0 + 10n \log_{10} \left(\frac{d}{d_0} \right) + X_\sigma \quad (6)$$

where:

- L_0 = path loss at reference distance d_0 (typically 1 m)
- n = path loss exponent
- X_σ = shadow fading (log-normal, std dev σ)

Environment	n	Loss per decade
Free space	2	20 dB
Urban cellular	2.7-3.5	27-35 dB
Indoor (LOS)	1.6-1.8	16-18 dB
Indoor (NLOS)	3-5	30-50 dB
Obstructed urban	4-6	40-60 dB

Table 1: Path loss exponents for different environments

Path Loss Exponent (n):**Key insight:** Obstacles dramatically increase path loss!

- $n = 3$: 30 dB per decade (vs. 20 dB free space)
- $n = 4$: 40 dB per decade (penetrating walls)

2.3 Shadow Fading and Multipath**Shadow Fading (Slow Fading):**

- Random variations due to obstacles
- Log-normal distribution: $X_\sigma \sim \mathcal{N}(0, \sigma^2)$
- Typical $\sigma = 4 - 12$ dB
- Changes slowly (seconds to minutes)

Multipath Fading (Fast Fading):

- Multiple signal paths arrive at receiver
- Constructive and destructive interference
- **Rayleigh fading:** No line-of-sight
- **Rician fading:** Dominant line-of-sight + multipath
- Changes rapidly (milliseconds)
- Can cause 20-30 dB signal variation!

Impact on Design:

- Need **fading margin** in link budget (10-20 dB)
- Diversity techniques (multiple antennas, frequencies, time slots)
- Adaptive modulation (change data rate based on channel quality)

2.4 Wireless Link Budget

Link budget equation:

$$P_r \text{ (dBm)} = P_t \text{ (dBm)} + G_t \text{ (dBi)} + G_r \text{ (dBi)} - L_{\text{path}} \text{ (dB)} - L_{\text{misc}} \text{ (dB)} \quad (7)$$

where:

- L_{misc} = cable loss, connector loss, polarization mismatch, etc.

Margin calculation:

$$\text{Margin} = P_r - S_{\text{min}} \quad (8)$$

where S_{min} = receiver sensitivity (minimum detectable signal)

Adequate margin:

- 10-15 dB: Acceptable for static links
- 20-30 dB: Needed for mobile/fading environments
- 40+ dB: Excellent, very reliable

3 Hands-On Activity 1: Wireless Link Budget Calculations

3.1 Problem 1: WiFi Coverage

Design a WiFi access point for 100m coverage:

- Frequency: 5 GHz
- Transmit power: 23 dBm (200 mW, regulatory limit)
- Antenna gains: 5 dBi (both)
- Environment: Indoor office ($n = 3$)
- Required SNR: 15 dB
- Noise floor: -95 dBm (in 20 MHz channel)
- Fading margin: 15 dB

Calculate:

1. Free space path loss at 100m
2. Actual path loss with $n = 3$ (use 1m reference)
3. Received power
4. Link margin

Solution

1. FSPL: $32.45 + 20 \log_{10}(5000) + 20 \log_{10}(0.1) = 86 \text{ dB}$
2. With $n = 3$: $L(100) = L(1) + 30 \log_{10}(100)$
 At 1m: $L(1) = 32.45 + 20 \log_{10}(5000) + 20 \log_{10}(0.001) = 46 \text{ dB}$
 At 100m: $L(100) = 46 + 30 \times 2 = 106 \text{ dB}$
3. $P_r = 23 + 5 + 5 - 106 = -73 \text{ dBm}$
4. $S_{\min} = -95 + 15 = -80 \text{ dBm}$ (noise + required SNR)
 Margin = $-73 - (-80) = 7 \text{ dB}$ (marginal with no fading margin!)
 With fading: Margin = $7 - 15 = -8 \text{ dB}$ (insufficient!)
Conclusion: Need higher transmit power, better antennas, or shorter range.

3.2 Problem 2: Cellular Tower

Cell tower covering 1 km radius:

- Frequency: 1.9 GHz (LTE)
- Transmit power: 43 dBm (20 W)
- Tower antenna gain: 17 dBi (directional, 120° sector)
- Phone antenna gain: 0 dBi
- Path loss exponent: $n = 3.5$
- Phone sensitivity: -110 dBm

Calculate received power and margin at cell edge (1 km).

Solution

$$\begin{aligned}
 L(1) &= 32.45 + 20 \log_{10}(1900) + 0 = 98 \text{ dB} \\
 L(1000) &= 98 + 35 \log_{10}(1000) = 98 + 105 = 203 \text{ dB} \\
 P_r &= 43 + 17 + 0 - 203 = -143 \text{ dBm}
 \end{aligned}$$

Margin = $-143 - (-110) = -33 \text{ dB}$ (no link!)

This shows why cell towers need high power and tall antennas!

4 Modulation as Channel Selection**4.1 Why Modulation?**

Baseband transmission problems:

- Antenna size $\approx \lambda/4$ (at 1 kHz: 75 km antenna!)

- All users share same spectrum (can't separate)
- Propagation poor at low frequencies

Solution: Modulation

- Shift baseband signal to higher carrier frequency
- Different users/channels use different carriers
- Practical antenna sizes
- Frequency Division Multiple Access (FDMA)

4.2 Modulation Schemes

1. Amplitude Modulation (AM):

$$s(t) = A[1 + m(t)] \cos(2\pi f_c t) \quad (9)$$

- Simple, but inefficient
- Sensitive to interference

2. Frequency Modulation (FM):

$$s(t) = A \cos \left(2\pi f_c t + 2\pi f_\Delta \int m(t) dt \right) \quad (10)$$

- More robust to noise
- Wider bandwidth

3. Phase Modulation (PM):

$$s(t) = A \cos(2\pi f_c t + \phi(t)) \quad (11)$$

- Most common in digital systems
- Efficient and robust

4.3 Digital Modulation: QAM

Quadrature Amplitude Modulation (QAM):

Key idea: Modulate both amplitude and phase

Transmitted signal:

$$s(t) = I(t) \cos(2\pi f_c t) - Q(t) \sin(2\pi f_c t) \quad (12)$$

where:

- $I(t)$ = in-phase component
- $Q(t)$ = quadrature component
- These are independent data streams!

Baseband representation:

$$s(t) = \text{Re}\{[I(t) + jQ(t)]e^{j2\pi f_c t}\} \quad (13)$$

Complex envelope: $z(t) = I(t) + jQ(t)$

4.4 Constellation Diagrams

Visualization: Plot symbol values in I-Q plane

BPSK (Binary Phase Shift Keying):

- 2 symbols: ± 1
- 1 bit/symbol
- Minimum distance: $d = 2$

QPSK (Quadrature PSK):

- 4 symbols at $\pm 1 \pm j$
- 2 bits/symbol
- Distance: $d = \sqrt{2}$

16-QAM:

- 16 symbols in 4×4 grid
- 4 bits/symbol
- Distance: smaller (requires higher SNR)

64-QAM, 256-QAM, 1024-QAM:

- Used in WiFi 6, 5G
- Higher spectral efficiency
- Require excellent SNR (>30 dB)

4.5 Symbol Error Rate for QAM

Approximate SER for square M -QAM:

$$P_s \approx 4 \left(1 - \frac{1}{\sqrt{M}} \right) Q \left(\sqrt{\frac{3}{M-1} \frac{E_s}{N_0}} \right) \quad (14)$$

For Gray coding: $P_b \approx P_s / \log_2(M)$

Required E_b/N_0 for BER = 10^{-6} :

Modulation	E_b/N_0 (dB)	Bits/symbol
BPSK	10.5	1
QPSK	10.5	2
16-QAM	14.5	4
64-QAM	18.5	6
256-QAM	23.5	8

Table 2: QAM modulation requirements for BER = 10^{-6}

Key insight: Each doubling of M requires ~ 3 -4 dB more SNR.

4.6 Adaptive Modulation

Modern systems adapt modulation to channel quality:

WiFi (802.11ac/ax) rate table:

MCS Index	Modulation	Code Rate	Data Rate (80 MHz)
0	BPSK	1/2	29 Mb/s
1	QPSK	1/2	58 Mb/s
2	QPSK	3/4	87.8 Mb/s
3	16-QAM	1/2	117 Mb/s
5	16-QAM	3/4	175.5 Mb/s
7	64-QAM	5/6	263 Mb/s
9	256-QAM	3/4	351 Mb/s
11	1024-QAM	3/4	468 Mb/s

Table 3: WiFi 802.11ac/ax MCS (Modulation and Coding Scheme) rates

Algorithm:

- Measure SNR at receiver
- Feedback to transmitter (or use reciprocity in TDD)
- Select highest MCS with acceptable error rate
- **Far from AP:** Low MCS (QPSK) $\rightarrow$ robust, slow
- **Near AP:** High MCS (256-QAM) $\rightarrow$ fragile, fast

5 Spectrum Allocation

5.1 Radio Frequency Spectrum

Frequency ranges:

Band	Frequency	Wavelength	Use
VLF	3-30 kHz	100-10 km	Submarine comm
LF	30-300 kHz	10-1 km	Navigation
MF	300-3000 kHz	1 km-100 m	AM radio
HF	3-30 MHz	100-10 m	Shortwave
VHF	30-300 MHz	10-1 m	FM radio, TV
UHF	300-3000 MHz	1 m-10 cm	TV, cellular, WiFi
SHF	3-30 GHz	10-1 cm	Satellite, 5G
EHF	30-300 GHz	1 cm-1 mm	mmWave 5G

Table 4: Radio frequency spectrum bands

5.2 Cellular Spectrum (US)

Licensed bands (exclusive use, auctioned by FCC):

4G LTE frequencies:

- **Band 2 (PCS):** 1850-1910 MHz (uplink), 1930-1990 MHz (downlink)
- **Band 4 (AWS):** 1710-1755 MHz (up), 2110-2155 MHz (down)
- **Band 12 (700 MHz):** 699-716 MHz (up), 729-746 MHz (down)
- **Band 66 (Extended AWS):** 1710-1780 MHz (up), 2110-2200 MHz (down)

5G frequencies:

- **Low band (Sub-6 GHz):** 600-900 MHz (wide coverage)
- **Mid band (C-band):** 3.7-3.98 GHz (capacity + coverage)
- **High band (mmWave):** 24-47 GHz (ultra-capacity, short range)

Frequency Division Duplexing (FDD):

- Separate uplink and downlink frequencies
- Both can operate simultaneously

Time Division Duplexing (TDD):

- Same frequency for up/down
- Alternate in time
- Used in some 5G bands

5.3 WiFi Spectrum (Unlicensed ISM Bands)

Anyone can use these bands (with power limits):

2.4 GHz band (2.400-2.4835 GHz):

- **Channels:** 1-11 (US), 1-13 (Europe)
- **Channel width:** 20 MHz (can overlap!)
- **Non-overlapping:** Channels 1, 6, 11 (in US)
- **Crowded:** Bluetooth, microwaves, cordless phones
- **Range:** Good (lower frequency)

5 GHz band (5.150-5.825 GHz):

- **Channels:** 36, 40, 44, 48, ... 165
- **Width:** 20, 40, 80, 160 MHz
- **Less crowded**

- **Range:** Shorter (higher frequency)
 - **Speed:** Higher (wider channels)
- 6 GHz band (5.925-7.125 GHz):** WiFi 6E

- **1200 MHz** of spectrum!
- Very wide channels (320 MHz)
- Brand new (2020)

Regulation:

- Max transmit power: 23 dBm (200 mW) typically
- Dynamic Frequency Selection (DFS) to avoid radar
- Transmit Power Control (TPC)

5.4 Other Wireless Systems

Bluetooth (2.4 GHz ISM):

- 79 channels, 1 MHz each
- Frequency hopping (1600 hops/sec)
- Short range (10-100 m)
- Low power

GPS:

- L1: 1575.42 MHz
- L2: 1227.60 MHz
- Receive-only (satellites transmit)
- Weak signal (-160 dBW at receiver!)

Satellite:

- C-band: 4-8 GHz (traditional)
- Ku-band: 12-18 GHz (DBS TV)
- Ka-band: 26-40 GHz (broadband)
- Starlink: Ku/Ka bands

LoRaWAN (IoT):

- 915 MHz (US), 868 MHz (EU)
- Ultra-long range (10+ km)
- Very low data rate (< 50 kbps)
- Low power

5.5 Spectrum Management

Licensed vs. Unlicensed:

Licensed	Unlicensed
Exclusive use	Shared
Expensive (auctions)	Free
Regulated power/quality	Best effort
Predictable performance	Interference possible
Example: Cellular	Example: WiFi, Bluetooth

Table 5: Licensed vs. unlicensed spectrum comparison

Spectrum is incredibly valuable!

- Recent US C-band auction (2021): **\$81 billion**
- Limited resource (like real estate)
- Tradeoff: coverage (low freq) vs. capacity (high freq)

6 Cellular Frequency Reuse

6.1 The Cellular Concept

Key idea: Divide coverage area into cells

- Each cell has a base station
- **Reuse frequencies** in non-adjacent cells
- Increases total system capacity

Without reuse:

- One transmitter covers entire city
- All users share single channel set
- **Very limited capacity**

With reuse:

- Many transmitters (cell towers)
- Same frequencies used in distant cells
- **Capacity multiplied by reuse factor**

6.2 Hexagonal Cell Pattern

Why hexagons?

- Approximate circular coverage
- Tile without gaps
- Geometric regularity

Cell cluster:

- Group of N cells using different frequencies
- Pattern repeats every N cells

Common patterns:

$N = 3$ (reuse every 3 cells):

- Aggressive reuse $\rightarrow$ more capacity
- Higher interference
- Requires good SNR

$N = 4$:

- Moderate reuse
- Balance capacity and quality

$N = 7$:

- Conservative reuse $\rightarrow$ lower interference
- Lower capacity per area
- Used in early cellular (AMPS)

6.3 Reuse Distance

Distance to nearest co-channel cell:

$$D = R\sqrt{3N} \tag{15}$$

where:

- D = reuse distance
- R = cell radius
- N = cluster size

Carrier-to-Interference Ratio (C/I):

Assuming 6 interfering cells in first tier:

$$\frac{C}{I} = \frac{1}{6} \left(\frac{D}{R} \right)^n = \frac{(3N)^{n/2}}{6} \quad (16)$$

where n = path loss exponent**For $n = 4$ (typical urban):**

- $N = 3$: $C/I = 18.6$ dB
- $N = 4$: $C/I = 24$ dB
- $N = 7$: $C/I = 31$ dB

Higher $N \rightarrow$ Better $C/I \rightarrow$ Lower capacity per cell**6.4 Modern Cellular: Frequency Reuse Factor = 1****Legacy systems (2G GSM, 3G):**

- Used $N = 3, 4$, or 7
- Different frequencies in adjacent cells
- Simple, but wastes spectrum

Modern systems (4G LTE, 5G NR):

- **Reuse factor = 1** (all cells use all frequencies!)
- How? Advanced techniques:
 - **Orthogonal Frequency Division Multiple Access (OFDMA)**
 - **Interference coordination** between base stations
 - **Beamforming** (directional antennas)
 - **Power control**
 - **Advanced receivers** (interference cancellation)

Result: Much higher spectral efficiency!**6.5 Sectoring****Problem:** Omnidirectional antennas waste power and create interference**Solution:** Divide cell into sectors

- Typically **3 sectors per cell** (120° each)
- Directional antennas (17-21 dBi gain)
- Each sector like a separate cell

Benefits:

- $3\times$ capacity (3 sectors)
- Lower interference (directional)
- Better link budget (antenna gain)

Towers typically have:

- 3 sectors (120° each)
- Multiple antennas per sector (MIMO)
- Multiple frequency bands

7 Summary & Key Takeaways

7.1 What We Learned

1. Wireless Architectures

- Transmitter: encoder $\rightarrow$ modulator $\rightarrow$ upconverter $\rightarrow$ PA $\rightarrow$ antenna
- Receiver: antenna $\rightarrow$ LNA $\rightarrow$ downconverter $\rightarrow$ demodulator $\rightarrow$ decoder
- LNA is critical (first stage dominates noise figure)

2. Propagation

- Free space path loss: 20 dB per decade of distance
- Real-world: path loss exponent $n = 2$ to 6
- Shadow fading: 4-12 dB variation
- Multipath fading: 20-30 dB variation (fast)

3. Link Budgets

- $P_r = P_t + G_t + G_r - L$
- Need 20-30 dB margin for fading
- Sensitivity limits coverage

4. Modulation and QAM

- Modulation shifts baseband to carrier (channel selection)
- QAM: independent I and Q streams
- Higher order QAM $\rightarrow$ better efficiency, needs higher SNR
- Adaptive modulation: change MCS based on channel

5. Spectrum Allocation

- Licensed: cellular (exclusive, expensive)
- Unlicensed: WiFi, Bluetooth (shared, free)
- 2.4 GHz: crowded, good range
- 5 GHz / 6 GHz: less crowded, shorter range

- mmWave (24+ GHz): huge capacity, very short range

6. Cellular Reuse

- Divide area into cells
- Reuse frequencies in distant cells
- Cluster size N : tradeoff capacity vs. interference
- Modern systems: $N = 1$ with advanced interference management
- Sectoring: $3\times$ capacity with directional antennas

7.2 Connection to Networking

Why wireless matters for networking:

- **Physical layer constraints affect everything above:**
 - Variable data rates (adaptive modulation)
 - Higher error rates than wired
 - Shared medium (requires MAC protocol)
 - Hidden/exposed terminal problems
- **WiFi design driven by physical layer:**
 - CSMA/CA (sense before transmit)
 - RTS/CTS (solve hidden terminal)
 - Channel selection (avoid interference)
- **Cellular optimized for mobility:**
 - Handoff between cells
 - Power control (don't waste battery)
 - Resource scheduling (fairness)
- **Tradeoffs permeate all layers:**
 - Power vs. performance
 - Range vs. data rate
 - Reliability vs. latency

7.3 Looking Ahead

Data Link Layer (Next):

- How to share the wireless channel (MAC protocols)
- Error detection and correction
- Frame formats
- WiFi (802.11) and Ethernet (802.3) in detail

Transport Layer:

- How TCP reacts to wireless losses
- Why TCP performs poorly on wireless
- Optimizations for mobile networks

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Additional Resources

Textbooks:

- Rappaport, “Wireless Communications: Principles and Practice”
- Goldsmith, “Wireless Communications”
- Tse & Viswanath, “Fundamentals of Wireless Communication”

Standards:

- IEEE 802.11 (WiFi)
- 3GPP specifications (LTE, 5G NR)
- ITU Radio Regulations

Tools:

- RF simulation: RF Explorer, inSSIDer
- Propagation: Radio Mobile, CloudRF
- SDR: GNU Radio, HackRF, USRP

Spectrum Maps:

- FCC frequency allocation chart
- National Table of Frequency Allocations